

Lab Seminar

(Fairness-aware Graph Learning)

Random Walk Approaches for Fair Node Embeddings



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Outline

❖ Introduction

- ❑ Graph representation learning
- ❑ Problems
- ❑ Motivation

❖ Fairness notation

❖ FairWalk

- ❑ A more diverse network neighborhood representation

❖ CrossWalk

- ❑ A bias random walk to cross group boundaries

❖ Conclusion

Graph: Definition

❖ What is a Graph?

- ❑ A general language for describing and analyzing **entities** with **relations/interactions**
- ❑ General view: a collection of objects (i.e. nodes), along with a set of interactions (i.e. edges) between pairs of these object
- ❑ Many types of data are graphs



Image credit: [Medium](#)
Social Networks

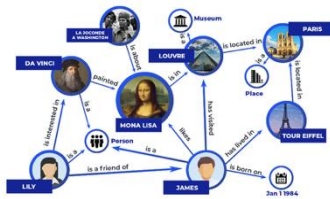


Image credit: [zilliz](#)
Knowledge Graph

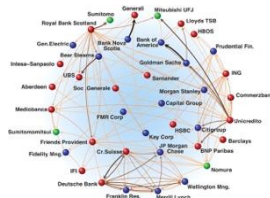


Image credit: [Science](#)
Economic Networks



Image credit: [missoulacurrent](#)
Internet



Image credit: [Lumen Learning](#)
Communication Networks

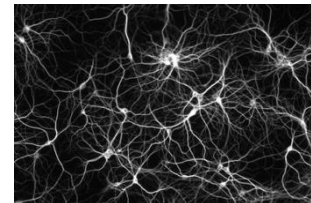
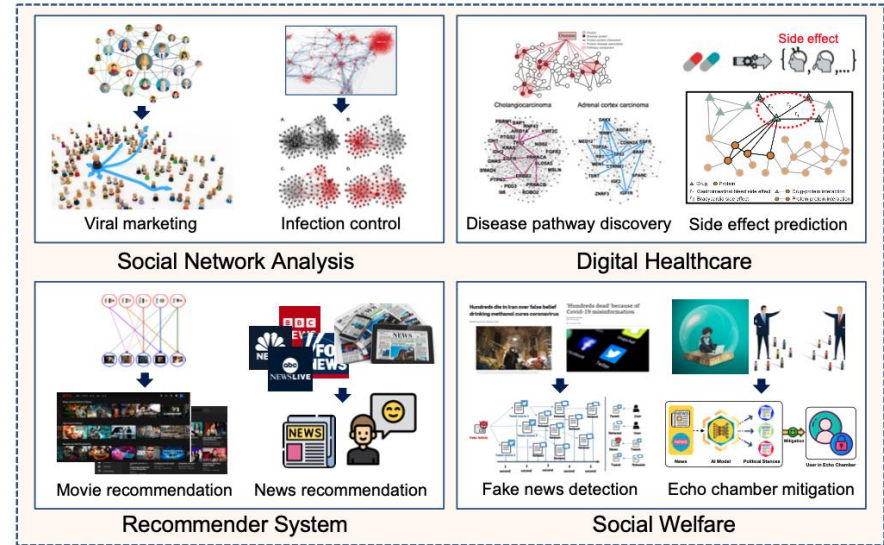
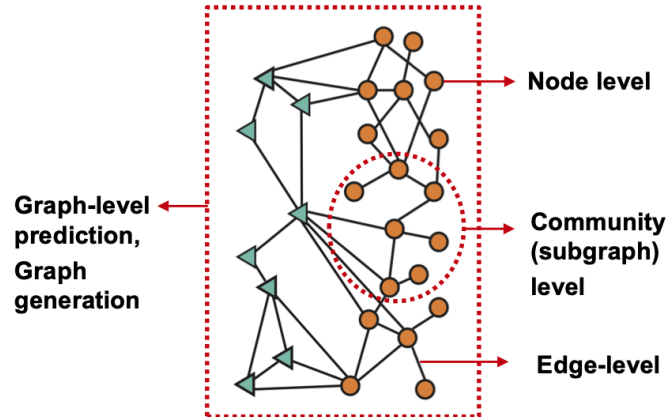


Image credit: [The Conversation](#)
Networks of Neurons

Graph: Application

❖ Different Types of Tasks

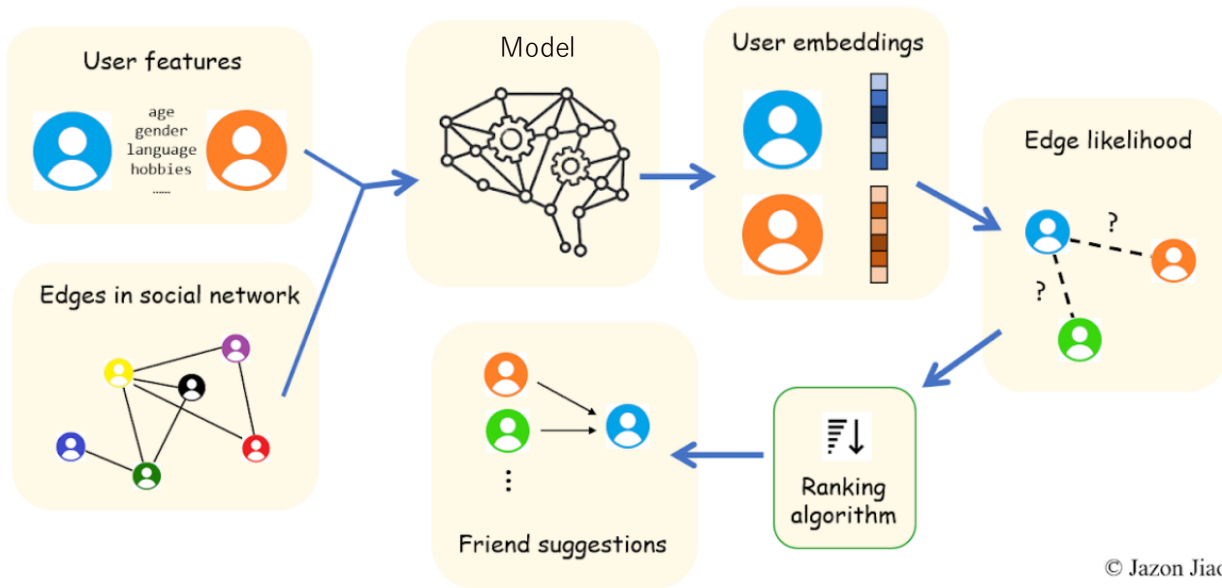
- ❑ Social Network Analysis
- ❑ Digital Healthcare
- ❑ Recommender System
- ❑ Social Welfare
- ❑ RAG, etc.



Graph: Application

❖ Pipeline for Friendship Recommendation

□ Graph-structured Data Integration ➡ Representation Learning ➡ Prediction Task



© Jazon Jiao

Fig. Recommender System

Recommender Systems

❖ Problems

- ❑ Minority communities get isolated and are not seen by others due to
 - ❑ (1) The echo chambers or
 - Users naturally connect with similar people—a concept called homophily
 - This creates dense clusters where information circulates rapidly but rarely "jumps" to other groups
 - ❑ (2) Information bubble effect
 - Algorithms tuned for "likelihood" often hide minority users from the majority
 - Because a match is deemed "statistically unlikely"



Echo Chambers

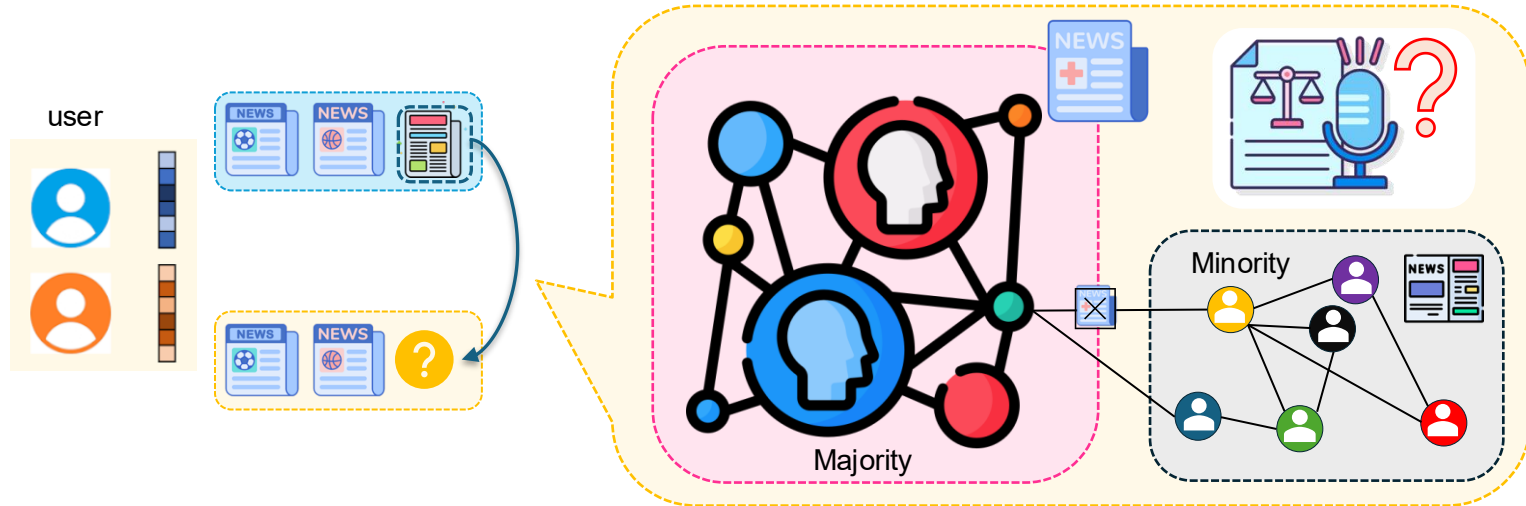


Information Bubble Effect

Problem - 1

❖ The Echo Chamber Effect (Cultural & Social)

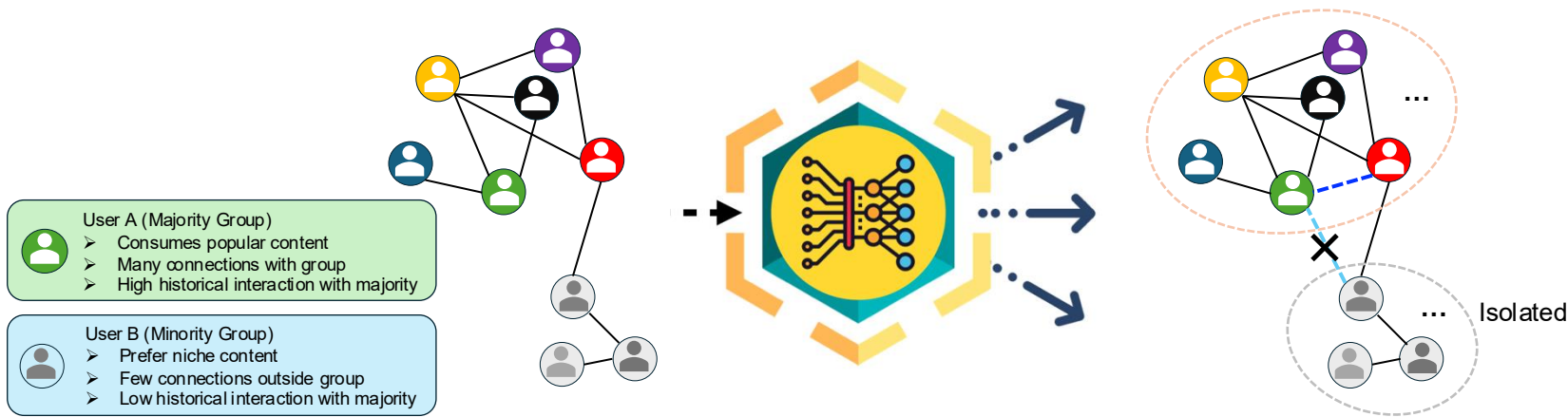
- ❑ Occur when individuals only encounter information or opinions that reflect and reinforce their own
- ❑ The user features often lead people to connect with those similar to them
- ❑ Result: Information circulates **rapidly within the group but rarely "jumps" to other clusters**



Problem - 2

❖ The Information Bubble (Algorithmic)

- ❑ If the data shows that
 - A minority user is 90% likely to interact with someone from their own small community and
 - Only 5% likely to interact with the majority
- ❑ The "Friend Suggestions" will almost exclusively feature the small community
- ❑ The minority community becomes **invisible** to the majority & the majority becomes an unreachable wall for the minority



Motivation

❖ Objective

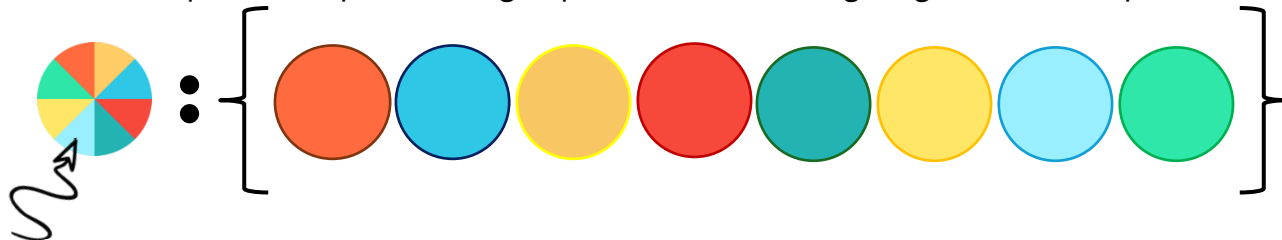
- ❑ To learn fair and high-utility node representations for downstream graph-based tasks

❖ Focus

- ❑ Achieving a synergy between high predictive accuracy and group-level fairness

❖ What We Need

- ❑ Diverse neighborhood samples that capture cross-group boundaries and long-range structural dependencies



Fairness-enhanced node embedding

❖ Learning Problem

- ❑ A low-dimensional fair embedding that encodes different properties of the graph structure along with the input features

Fairness: Definition

❖ Principle

- ❑ Lack of favoritism from one side or another

❖ Various Definition of Fairness



Fairness definition

Two sides

Group fairness

Two demographic groups

Individual fairness

Two data points

Counterfactual fairness

A data point and its counterfactual version

Difference principle

Two groups of points with different utility



Images credit: [idiro](#)

Fairness Notation

❖ Group Fairness

- ❑ Defined upon sensitive subgroups
- ❑ Subgroups are divided by
 - Features protected **by law** to avoid being abused (e.g., race and gender)
 - Features that **users** are usually **unwilling to share** (e.g., occupation, age)
 - Usually, sensitive features are binary

“Generally speaking, **group fairness** requires that the algorithm should **not yield discriminatory predictions or decisions against individuals from any specific sensitive subgroup.**”



Images credit: [idiro](#)

Group Fairness: Definition

❖ Fairness Metrics

- Statistical Parity (a.k.a Demographic Parity) – SP/DP

➤ **Equal acceptance rate** : $\Delta SP = |P(\hat{Y} = 1 | S = 0) - P(\hat{Y} = 1 | S = 1)|$

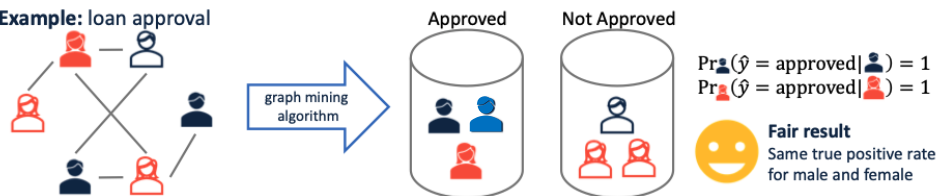
• **Example: loan approval**



- Equal Opportunity - EO

➤ **Equal true positive rate** : $\Delta EO = |P(\hat{Y} = 1 | Y = 1, S = 0) - P(\hat{Y} = 1 | Y = 1, S = 1)|$

• **Example: loan approval**





FairWalk: Towards Fair Graph Embedding

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CISPA Helmholtz Center for Information Security

IJCAI-19

Motivation

❖ Fairness of node2vec

- ❑ Specific task: **friendship recommender system**
- ❑ The system recommends a candidate from the same group
- ❑ Fig. 1 shows the bias in the output of node2vec are similar with original network

➤ **Mirror the gap between minorities and majorities**

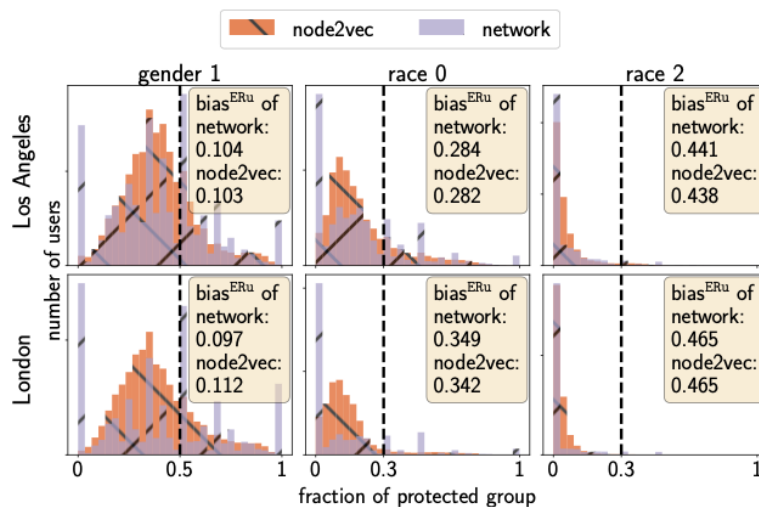


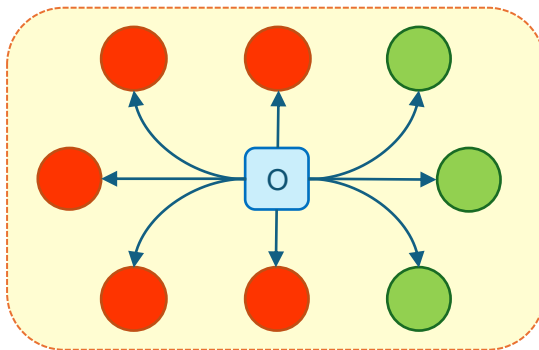
Figure 1: z-share distributions of node2vec and original network. The vertical line shows the *fair fraction* (0.5 and 0.3)

A More Diverse Network Neighborhood Representation

❖ A Modified Version of Random Walk (node2vec)

- ❑ Change random walk procedure from **random to biased way**
- ❑ **Partition neighbors into groups** based on their sensitive attribute values
- ❑ **Give each group the same probability** of being chosen regardless of their sizes
- ❑ Then a random node from the chosen group is selected for the jump

Idea is simple:



Probability of next walk:

➤ **Red** : $5/8 \rightarrow 1/2$

➤ **Green** : $3/8 \rightarrow 1/2$

- Not only does it result in **higher network neighborhood diversity** with respect to sensitive attribute
- But also **minorities appear more frequently** in the traces

Experiment

❖ Setup

- ❑ Data : Instagram API, 2016
- ❑ Sensitive Attribute : Race and Gender
- ❑ Task : Friend Recommendation

❖ Resulting Traces

- ❑ Fig. 2 shows that minorities appear more often in the fair random walk
- ❑ This provides more network neighborhood data for minorities
- ❑ Which enables the neural network give more importance to obtain their best vector representation

	LA	London
No. users	82,607	53,902
No. social links	482,305	165,184
gender 0	62.6%	62.3%
gender 1	37.4%	37.7%
race 0	21.9%	15.9%
race 1	72.2%	80.7%
race 2	5.9%	3.4%

Table 2: Statistics of both datasets.

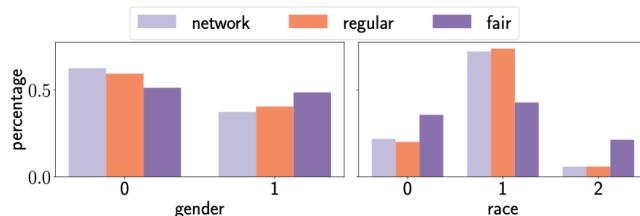


Figure 2: Ratio of each gender and race in the original network and regular and fair random walk traces in Los Angeles dataset

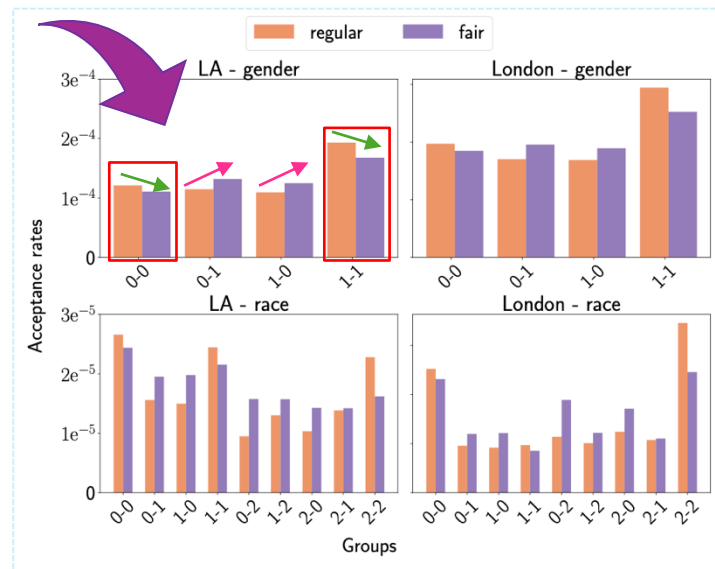
Experiment

❖ Statistical Imparity

- ❑ Fig. 3 shows the **accepted rates** for both the regular & FairWalk
 - FairWalk increases the probability of the under-represented groups being recommended to a very large extent
 - The probabilities of **same gender and same race** friendship recommendations are always **reduced**
 - The probabilities of **diverse friendships** are always **increased**
- ❑ Table. 3 shows bias^{SI} compared to node2vec for both cities
 - Fairwalk reduces $\text{bias}^{\text{SI}}_{\text{gender}}$ by 61% and $\text{bias}^{\text{SI}}_{\text{race}}$ by 68% for LA
 - For London, reduce $\text{bias}^{\text{SI}}_{\text{gender}}$ by 91% and $\text{bias}^{\text{SI}}_{\text{race}}$ by 61%

		LA		London	
		gender	race	gender	race
ER_g	regular	$1.3e^{10}$	$2.5e^7$	$6.5e^9$	$2.4e^7$
	fair	$0.8e^{10}$	$1.9e^7$	$4.8e^9$	$1.9e^7$
SI	regular	$4.7e^{-9}$	$1.4e^{-12}$	$1.1e^{-8}$	$7.1e^{-11}$
	fair	$1.7e^{-9}$	$0.4e^{-12}$	$0.2e^{-8}$	$2.8e^{-11}$

Table 3: bias^{SI} and $\text{bias}^{\text{ER}_g}$ for both cities (lower, the better)



Experiment

❖ Equality of Representation

- Minority communities get isolated and are not seen by others due to echo chambers or the information bubble effect
- To promote recommendations where all groups are equally represented, bias by **Equality of Representation** is defined

In two variants:

- bias^{ERg} at the network level (structural output imbalance)
- bias^{ERu} at the user level (user experience imbalance)

□ Group Level: Fig. 4

- Decrease the share for the over-represented groups [1]
- Increase the share for most underrepresented groups [2]
- Not increase the representation anymore since they are already very strongly connected

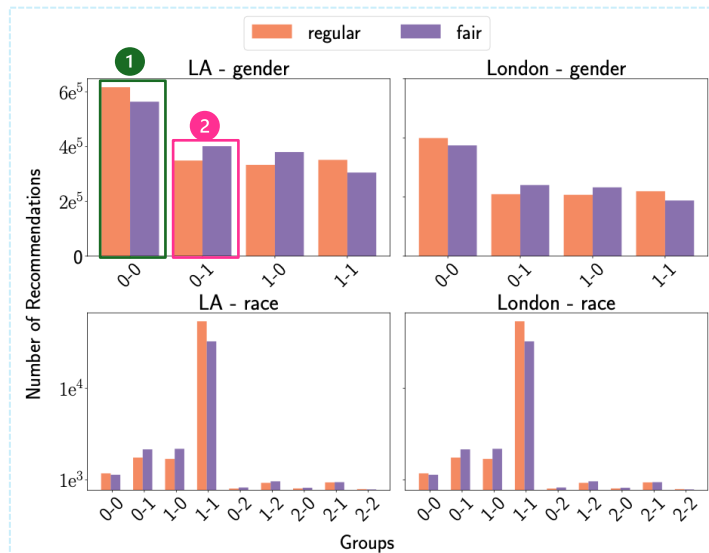


Figure 4: Number of recommended users pairs from each group.

Experiment

❖ Equality of Representation

- Minority communities get isolated and are not seen by others due to echo chambers or the information bubble effect
- To promote recommendations where all groups are equally represented, bias by **Equality of Representation** is defined

In two variants:

- bias^{ERG} at the network level (structural output imbalance)
- bias^{ERu} at the user level (user experience imbalance)

□ User Level: Fig. 5

- Consistently **shifts** these distributions **toward the ideal fair fraction**
- Showing reduced echo chambers and improved minority visibility

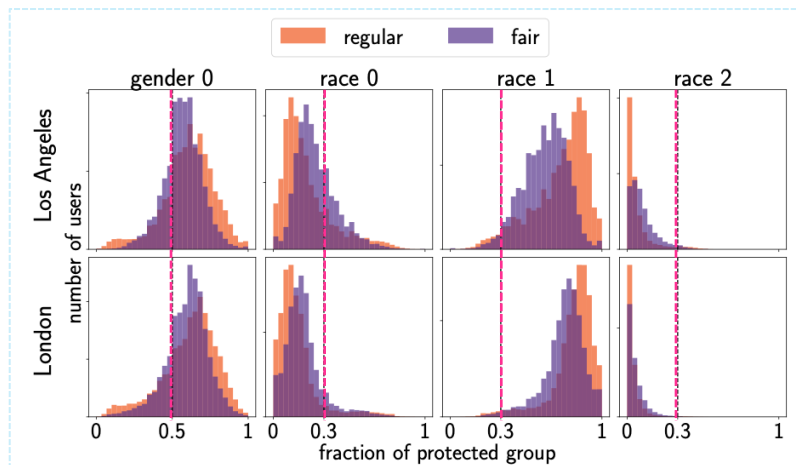


Figure 5: z -share distributions of node2vec and Fairwalk. The vertical line shows the fair fraction.



CrossWalk: Fairness-Enhanced Node Representation Learning

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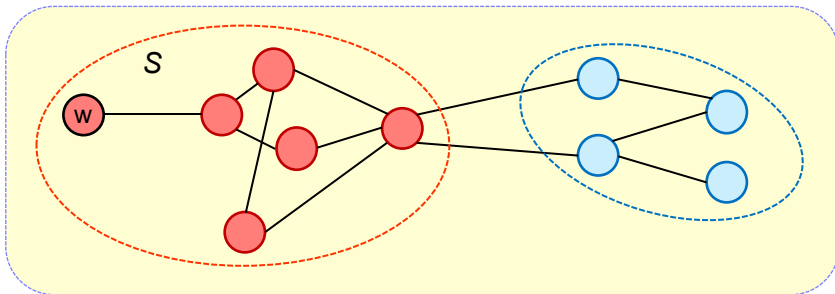
⁶The Alan Turing Institute

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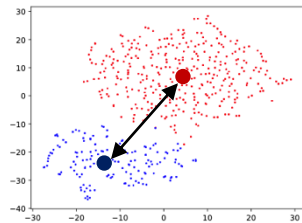
AAAI-22

Motivation

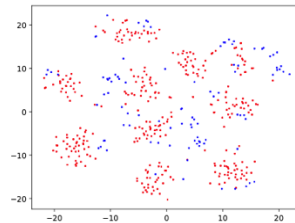
- ❖ **FairWalk: Fail to capture the neighborhoods located in more than one-hop away information**



- ❖ **Need to manipulate random walk process in more detail**



(a) Synthetic Layered Dataset
DeepWalk

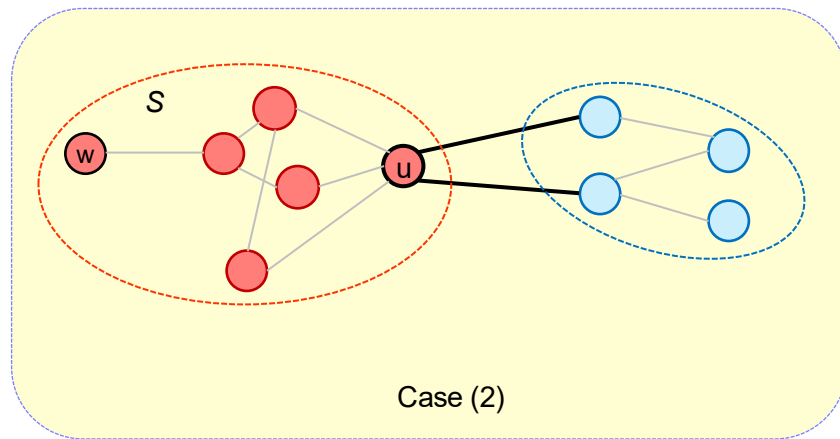
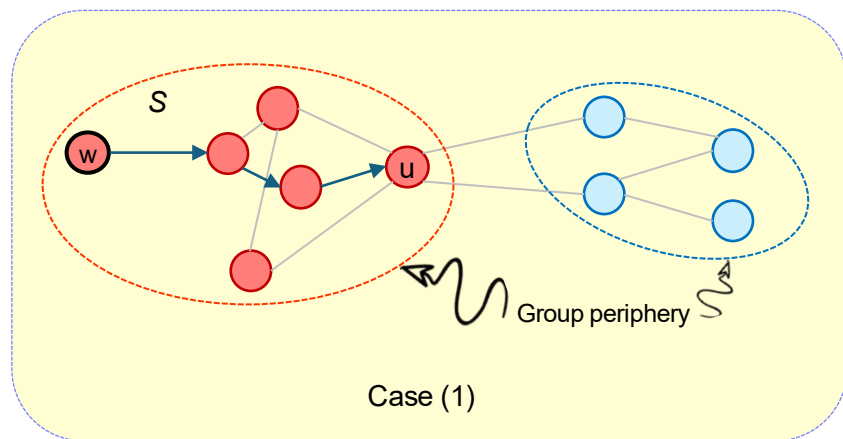


(c) Rice-Facebook Dataset
DeepWalk

Method

❖ Key Idea: To Bias Random Walks to Cross Group Boundaries

- ❑ Upweighting edges
 - (1) Closer to the groups' peripheries
 - (2) Connecting different groups in the network



Methodology

❖ CrossWalk: random walks initiated from a particular group

- ❑ Will be pulled toward the group boundaries
- ❑ Have a higher probability of
 - Crossing groups' peripheries
 - Visiting nodes from other groups in the network

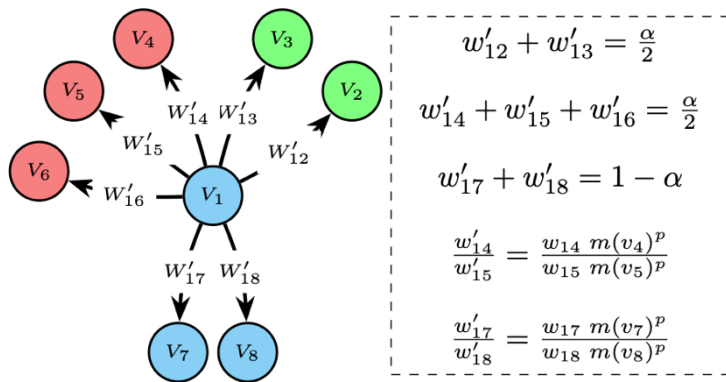


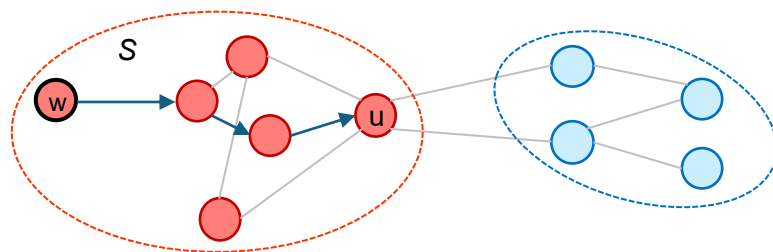
Figure 1: Illustrating CrossWalk. Different colors represent nodes from different groups. Our method upweights edges which are (1) closer to the groups' peripheries or (2) connecting different groups in the network.

Methodology

❖ Biased Towards Group Boundaries

- A measure of **proximity** to other groups in the graph
- The expected number of times nodes from other groups are visited in r -random walks of length- d rooted at v

$$m(v) = \frac{\sum_{j \in [r]} \sum_{u \in \mathcal{W}_v^j} \mathbb{I}[l_v \neq l_u]}{r \times d}.$$



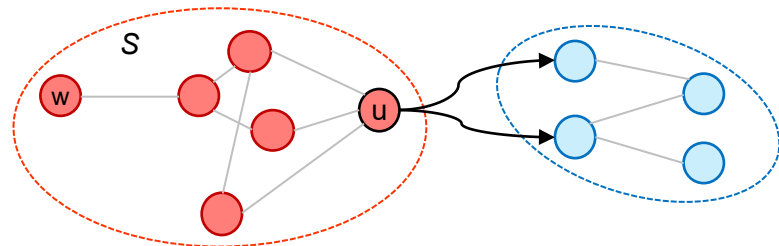
- Every node has $m(v)$, but following has a higher value
 - Nodes that are closer to group boundaries
 - Nodes that have a larger number of nodes with a different label in their close proximity
- **Assigning larger weights to the edges connecting nodes with larger proximity values**
- **Can capture more distant neighborhood information due to considering random walk path**

Methodology

❖ Bias towards Other Groups

- ❑ To bias the stochastic edge traversal procedure towards visiting nodes from other groups
- ❑ Upweight the edges connecting different groups in the graph

$$w'_{vu} = \begin{cases} w_{vu}(1 - \alpha) \times \frac{m(u)^p}{\sum_{z \in N_v} w_{vz} m(z)^p} & \text{if } l_v = l_u \text{ (same group)} \\ w_{vu}\alpha \times \frac{m(u)^p}{|R_v| \sum_{z \in N_v^c} w_{vz} m(z)^p} & \text{if } l_v \neq l_u = c. \text{ (different group)} \end{cases}$$



- α : control strengths to the other group (The higher, the more powerful)
- p : control strengths to the group boundaries (The higher, the more powerful)

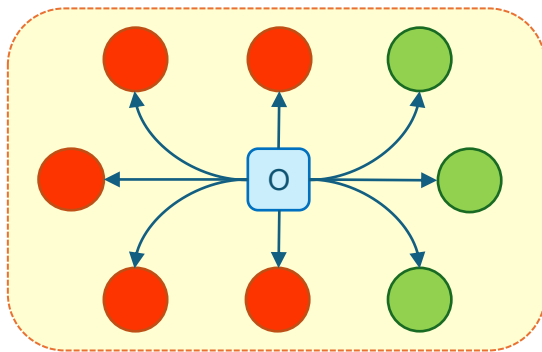
(Recap)

FairWalk: A More Diverse Network Neighborhood Representation

❖ A Modified Version of Random Walk (node2vec)

- ❑ Change random walk procedure from **random to biased way**
- ❑ **Partition neighbors into groups** based on their sensitive attribute values
- ❑ **Give each group the same probability** of being chosen regardless of their sizes
- ❑ Then a random node from the chosen group is selected for the jump

Idea is simple:



Probability of next walk:

➤ **Red** : $5/8 \rightarrow 1/2$

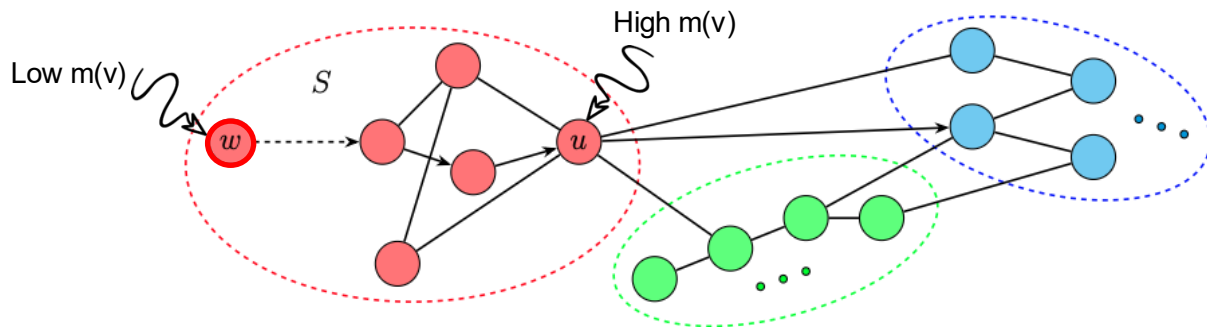
➤ **Green** : $3/8 \rightarrow 1/2$

- Not only does it result in **higher network neighborhood diversity** with respect to sensitive attribute
- But also **minorities appear more frequently** in the traces

Comparison with FairWalk

❖ (1) Inner-group Connections

CrossWalk	FairWalk
Carefully reweight by assigning larger weights to edges that are closer to group peripheries	Not bias random walks initiated within a group towards visiting nodes at the group's peripheries

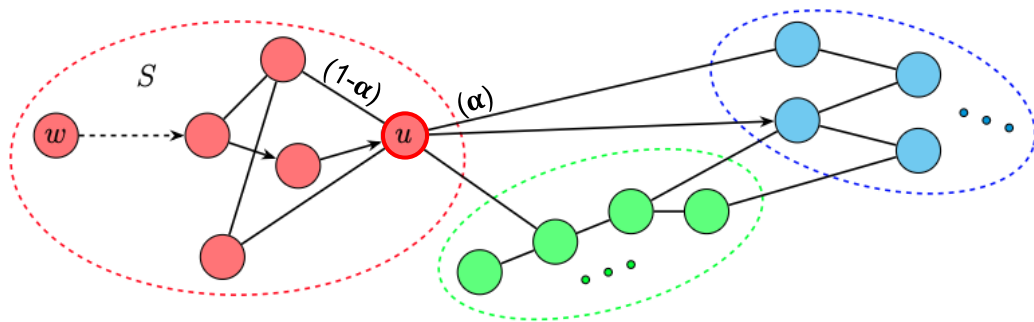


FairWalk fails to enhance fairness in graphs where the **majority of nodes are more than one hop** away from group peripheries

Comparison with FairWalk

❖ (2) Stay-vs-leave the Group at Boundary Nodes

CrossWalk	FairWalk
Carefully controls cross-group transitions by enforcing a constant probability $1-\alpha$ of staying within the same group	Splits probability across all groups at boundary nodes, causing minority nodes to immediately lose group cohesion when connected to many other groups



$u \rightarrow \text{own-group node} \rightarrow \text{own-group node} \rightarrow u \rightarrow \text{other group (Or)}$
 $u \rightarrow \text{other group} \rightarrow \text{other group} \rightarrow \text{returns to } u \text{ with non-trivial probability}$

$u \rightarrow \text{other group} \rightarrow \text{other group} \rightarrow \text{never returns to group } c$

FairWalk fails to **preserve minority group structure** when boundary nodes are connected to many external groups

Experiments

❖ Setup

- ❑ Use DeepWalk embedding
- ❑ For influence maximization, use greedy and adversarial graph embedding as baseline

❖ Influence Maximization

- ❑ Fig.4 (a) & (b)

CrossWalk results in a much **larger decrease in disparity**

than FairWalk and Adversarial embedding,

with a very **small decrease in the total influence**

- ❑ On Twitter dataset that FairWalk increases the disparity compared to DeepWalk

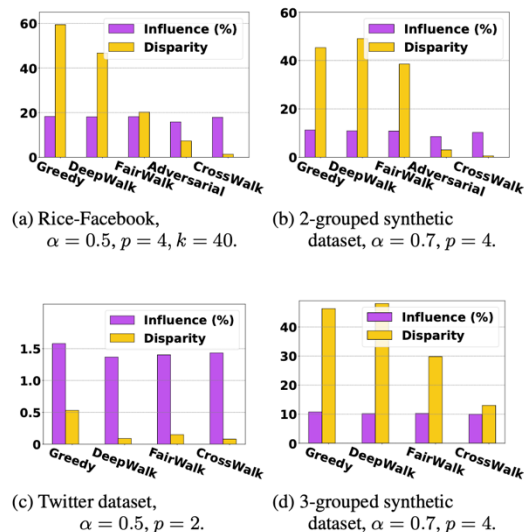


Figure 4: Influence Maximization

Experiments

❖ Influence Maximization

- Apply CrossWalk to Node2Vec

❖ Link Prediction

- The superiority of CrossWalk over FairWalk in reducing disparity with a slight decrease in accuracy

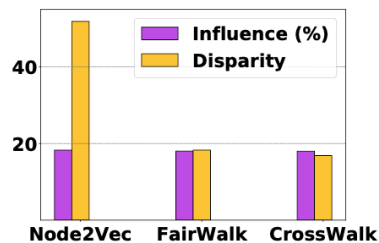


Figure 5: Influence Maximization - CrossWalk and FairWalk on Node2Vec

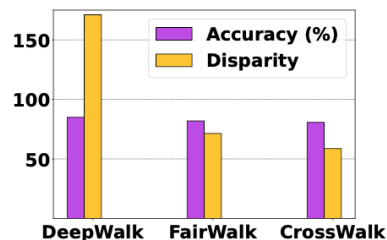
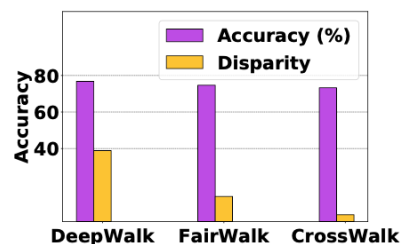
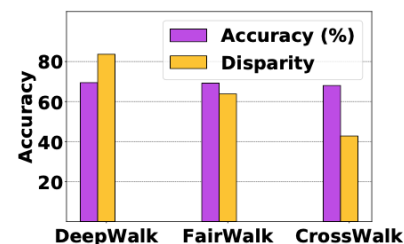


Figure 6: Node Classification - Rice-Facebook dataset. $\alpha = 0.5, p = 1$.



(a) Rice-Facebook Dataset, $\alpha = 0.5, p = 2$.



(b) Twitter Dataset, $\alpha = 0.5, p = 2$

Figure 7: Link Prediction

Conclusion

❖ Introduction & Problem Statement

- ❑ Fundamentals & Graph Applications
- ❑ The Fairness Problem: Standard graph algorithms often isolate minority communities through two effects
 - The Echo Chamber (Social/Cultural) & The Information Bubble (Algorithmic)

❖ Defining Fairness

- ❑ The core principle of fairness is the **lack of favoritism**
- ❑ **Group Fairness**—ensuring that sensitive subgroups (defined by race, gender, etc.) do not receive discriminatory predictions
 - Statistical Parity (SP): Do groups have equal acceptance rates?
 - Equal Opportunity (EO): Do groups have the same true positive rate?

❖ Algorithm 1: FairWalk (IJCAI-19)

- ❑ Motivation: If the original network is segregated, the resulting recommendations will be too
- ❑ Strategy: It **partitions neighbors into groups** based on sensitive attributes. It gives each group an **equal probability** of being chosen, regardless of size

❖ Algorithm 2: CrossWalk (AAAI-22)

- ❑ Motivation: FairWalk struggles to capture information more than one hop away
- ❑ Strategy: Upweighting edges - (1) Closer to the groups' peripheries & (2) Connecting different groups in the network

Thank You!



"Output is conditional on INPUT."